Megalencephaly–Polymicrogyria–Polydactyly–Hydrocephalus Syndrome 3 (MPPH3)

A Comprehensive Disease Characterization Report

Causal gene: *CCND2* (cyclin D2). MPPH3 is the *CCND2*-caused subtype of the megalencephaly–polymicrogyria–polydactyly–hydrocephalus (MPPH) spectrum. It is caused by de novo, activating (protein-stabilizing) heterozygous variants in *CCND2*, which encodes cyclin D2, a G1-phase cell-cycle regulator that partners with CDK4/CDK6. Note: the disease name is a serial number ("...syndrome 3") that does not itself name the gene; per the provided identifiers (MONDO:0014408, OMIM 615938), this report concerns the ***CCND2*** form, distinct from MPPH1 (*PIK3R2*) and MPPH2 (*AKT3*).

> **Evidence-source note:** Because MPPH3 is an ultra-rare Mendelian disorder (<100 total MPPH cases of all genotypes reported), the evidence base is dominated by human case reports/case series and foundational functional genetics, supplemented by mouse models (knockout; in utero electroporation) and in vitro biochemistry. There are no large registries, GWAS, or omics cohorts specific to MPPH3.

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1. Disease Information

Overview. MPPH3 is a rare congenital brain-overgrowth (megalencephaly) and malformation-of-cortical-development syndrome. The cardinal tetrad is **megalencephaly** (brain overgrowth), **polymicrogyria** (classically bilateral perisylvian), **postaxial polydactyly**, and **ventriculomegaly/hydrocephalus**. It is one end of a continuum of PI3K–AKT–mTOR "megalencephaly syndromes" that also includes the megalencephaly–capillary malformation syndrome (MCAP).

> "Megalencephaly-polymicrogyria-polydactyly-hydrocephalus syndrome (MPPH)... is characterized by progressive megalencephaly, cortical brain malformations, and distal limb anomalies." (P 38700464)

Key identifiers. - **OMIM:** 615938 (MPPH3). The historical MPPH locus entry is OMIM 603387. - **MONDO:** MONDO:0014408 - **Orphanet:** MPPH spectrum = ORPHA:83473 (megalencephaly–polymicrogyria–postaxial polydactyly–hydrocephalus syndrome) - **Gene:** *CCND2* — HGNC:1583; NCBI Gene 894; Ensembl ENSG00000118971; UniProt P30279 (G1/S-specific cyclin-D2); chromosome 12p13.32. - **MeSH:** closest terms — Megalencephaly; Polymicrogyria; Malformations of Cortical Development; Hydrocephalus. - **ICD-11:** LD90.0 (megalencephaly) / LA05 (congenital malformations of brain) are the best-fit umbrella codes; no MPPH3-specific code. **ICD-10:** Q04.3 / Q04.8 (other congenital brain malformations).

Synonyms / alternative names: megalencephaly-polymicrogyria-polydactyly-hydrocephalus syndrome type 3; CCND2 megalencephaly-polymicrogyria-postaxial polydactyly-hydrocephalus syndrome; MPPH caused by mutation in CCND2.

Data source type: Aggregated from individual-patient case reports/series and functional-genetics studies; not derived from EHR or disease-level omics aggregation.

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2. Etiology

Primary cause — genetic. De novo heterozygous **gain-of-function** (protein-stabilizing) variants in *CCND2*.

> "individuals with MPPH lacking upstream PI3K-AKT pathway mutations carry de novo mutations in CCND2 (encoding cyclin D2) that are clustered around a residue that can be phosphorylated by glycogen synthase kinase 3β (GSK-3β). Mutant CCND2 was resistant to proteasomal degradation in vitro compared to wild-type CCND2." (Mirzaa et al. 2014, P 24705253)

Genetic risk factors. The disease-defining lesions are the de novo *CCND2* variants themselves (see §4). There are no established common susceptibility loci or modifier genes; disease is monogenic/Mendelian with essentially complete penetrance for carriers of the stabilizing variants.

Environmental risk factors. None established. The causal mutations are de novo germline (or rarely postzygotic mosaic) events; no toxin, infection, maternal exposure, or lifestyle factor is implicated. Advanced parental age is a general (non-specific) correlate of de novo mutation rate but has not been specifically demonstrated for *CCND2*.

Protective factors. None identified. Conceptually, enhancing GSK-3β-dependent cyclin D2 degradation or inhibiting upstream PI3K–AKT signaling would be predicted to be "protective" (basis for targeted-therapy rationale), but no genetic protective allele is known.

Gene–environment interactions. Not applicable/none documented for this monogenic developmental disorder.

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3. Phenotypes

All frequencies are qualitative/approximate given small case numbers. Onset is **prenatal/congenital** for structural features; neurological features (seizures, delay) manifest in infancy. Course is generally **stable-structural but with progressive megalencephaly and often progressive ventriculomegaly**; neurodevelopmental impairment is lifelong.

Phenotype	Type	HPO term	Frequency	Onset / course	QoL impact
Megalencephaly / macrocephaly (HC > +2 to +3 SD)	physical sign / imaging	HP:0001355 Megalencephaly; HP:0000256 Macrocephaly	Near-universal (defining)	Congenital, progressive	High — mechanical, cosmetic, comorbidity driver
Polymicrogyria (bilateral perisylvian typical)	imaging / structural	HP:0002126 Polymicrogyria	Very frequent (defining)	Prenatal	High — drives epilepsy & delay
Ventriculomegaly / hydrocephalus	imaging / clinical	HP:0002119 Ventriculomegaly; HP:0000238 Hydrocephalus	Frequent (defining)	Pre/ perinatal, often progressive	High — may need CSF diversion
Postaxial polydactyly	physical malformation	HP:0100259 Postaxial polydactyly	Variable; often absent in *CCND2* cases	Congenital	Low–moderate (surgically correctable)
Developmental delay / intellectual disability	behavioral / cognitive	HP:0001263 Global developmental delay; HP:0001249 Intellectual disability	Near-universal	Infancy, lifelong	Severe
Epilepsy / seizures (incl. infantile spasms)	neurological sign	HP:0001250 Seizure; HP:0012469 Infantile spasms	Common	Infancy–childhood	Severe
Hypotonia	neurological sign	HP:0001252 Hypotonia	Common	Infancy	Moderate–severe
	imaging	HP:0002707 Cerebellar		Congenital	Variable

Phenotype	Type	HPO term	Frequency	Onset / course	QoL impact
Cerebellar tonsillar ectopia / Chiari-like		tonsillar herniation-related	Reported in MPPH spectrum		
Distinctive facies / prominent forehead (frontal bossing, hypertelorism, low nasal bridge)	physical	HP:0011220 Prominent forehead; HP:0002007 Frontal bossing; HP:0000316 Hypertelorism	Frequent	Congenital	Low
Thick corpus callosum	imaging / structural	HP:0007074 Thick corpus callosum	Recognized MPPH feature	Congenital	Variable
Severe intellectual disability / absent speech	cognitive / behavioral	HP:0010864 Severe intellectual disability; HP:0001344 Absent speech	Common	Childhood, lifelong	Severe
Syndactyly	physical malformation	HP:0001159 Syndactyly	Variable	Congenital	Low
Intraventricular hemorrhage (reported)	clinical	HP:0030746 Intracerebral hemorrhage	Reported (CCND2 p.Thr280Ile case)	Perinatal	High when present

> **Authoritative HPO annotation set (Monarch/HPO for MONDO:0014408, 12 curated terms):** HP:0001355 Megalencephaly · HP:0000256 Macrocephaly · HP:0002126 Polymicrogyria · HP:0002119 Ventriculomegaly · HP:0000238 Hydrocephalus · HP:0007074 Thick corpus callosum · HP:0001263 Global developmental delay · HP:0010864 Severe

> intellectual disability · HP:0001344 Absent speech · HP:0001162 Postaxial hand polydactyly · HP:0001159 Syndactyly · HP:0011220 Prominent forehead. (Explicit frequency qualifiers not provided by the source.)
>
> A reported *CCND2* (p.Thr280Ile) case had "ventriculomegaly, grade 3 IVH, bilateral polymicrogyria, and restricted diffusion in the caudate nuclei prenatally. No polydactyly was observed. The patient developed infantile spasms at age 5 months... By age 2 years, the patient continued to have significant developmental delays, including having poor tone and being nonverbal." (P 39395260)
> — illustrating that **polydactyly is variable/often absent** while neurodevelopmental and CSF phenotypes dominate.

Quality of life. Dominated by severe/profound neurodevelopmental disability, drug-resistant epilepsy in a subset, and complications of hydrocephalus; no disease-specific QoL instruments reported.

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4. Genetic / Molecular Information

Causal gene. *CCND2* (cyclin D2); HGNC:1583; NCBI Gene 894; UniProt P30279; chr12p13.32; OMIM gene 123833; disease OMIM 615938.

Pathogenic variants. - **Location/clustering:** De novo variants cluster in the **terminal (last) exon** around the C-terminal **GSK-3β phosphodegron (Thr280, "T280")**. Examples: p.Thr280Ala, p.Thr280Ile (c.839C>T), and nearby residues; splice/truncating changes that remove the C-terminal degron also occur. - **ClinVar-cataloged MPPH3 alleles (transcript NM_001759.4; accessed 2026-11):** c.838A>C **p.(Thr280Pro)** [Pathogenic/Likely pathogenic]; c.802G>T **p.(Gly268Ter)** [Likely pathogenic]; c.812C>A **p.(Ser271Ter)** [Likely pathogenic]; c.829C>T **p.(Gln277Ter)** [Pathogenic/Likely pathogenic], all annotated to MPPH3/CCND2-related disorder and located in the final exon (≈codons 268–284). **Key nuance:** C-terminal **nonsense/frameshift** variants escape nonsense-mediated decay and delete the phosphodegron, yielding the same stabilized, degradation-resistant protein as the Thr280 missense alleles (gain of function). In contrast, **proximal truncating** variants (e.g., c.416_419dup p.Leu141fs, c.643C>T p.Gln215Ter) are true loss-of-function and cause the reciprocal **microcephaly** phenotype

( ).
P 34087052
Of 94 P/LP CCND2 ClinVar records, most are large 12p copy-number changes unrelated to the single-gene MPPH3 mechanism.

> "Megalencephaly-associated CCND2 variants are localized to the terminal exon and result in accumulation of degradation-resistant protein." ( )
> P 34087052
> - **Variant type/class:** Predominantly **missense** at/near Thr280, plus splice-site variants affecting the terminal exon; all converge on **loss of the degron → protein stabilization**. - **Classification (ACMG/AMP):** Recurrent distal/Thr280 variants are classified **pathogenic/likely pathogenic** (de novo occurrence PS2, functional evidence PS3, clustering/absence from controls). - **Allele frequency:** Absent from population databases (gnomAD) — de novo, not polymorphic (ACMG PM2). - **Gene constraint (gnomAD, computational):** CCND2 is highly dosage-sensitive — **pLI = 0.999**, **LOEUF (oe_lof upper) = 0.39**, observed/expected LoF = 6/30.2, LoF Z = 3.74; **missense Z = 3.06** (obs/exp 235/373). This strong intolerance to both LoF and missense variation is the population-genetics correlate of the bidirectional "inverse phenotype" (LoF → microcephaly; degron-disrupting gain of function → megalencephaly/MPPH3). - **Protein-level annotation (UniProt P30279, G1/S-specific cyclin-D2, 289 aa):** The pathogenic cluster maps to the intrinsically **disordered C-terminal region (begins ≈residue 264)** that carries the phosphodegron. UniProt independently annotates **Thr280 as a phosphothreonine** and the adjacent **Ser271 as a phosphoserine** modified residue — directly corroborating that the GSK-3β-targeted degron (Thr280) and the codon-268–284 variant hotspot sit within the degradation-control module. Loss of this phosphothreonine (missense or C-terminal truncation) is therefore the structural basis for the stabilized gain-of-function protein. - **Germline vs somatic:** Typically **de novo germline**; postzygotic mosaicism occurs across the broader PI3K–AKT megalencephaly spectrum (relevant for recurrence risk and tissue-specific testing). - **Functional consequence: Gain of function** via increased protein stability/abundance (not altered intrinsic enzymatic activity of the complex).

> Reciprocal (inverse) phenotype: **proximal loss-of-function** *CCND2* variants cause **microcephaly, short stature, developmental delay** — "distinct classes of CCND2 variants are associated with reciprocal effects on human brain growth (microcephaly and megalencephaly due to possible loss or gain of protein function, respectively)." (P 34087052)

Modifier genes. None established; phenotypic variability (e.g., presence/absence of polydactyly, seizure severity) is unexplained and may reflect variant identity and stochastic/developmental factors.

Epigenetic / chromosomal. No recurrent methylation signature or chromosomal abnormality defines MPPH3; it is a single-nucleotide/small-variant disorder. Chromosomal microarray is typically normal (e.g., P 38700464 reported no CNVs).

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5. Environmental Information

Not applicable. MPPH3 is a monogenic developmental disorder caused by de novo mutation. No environmental toxin, radiation, lifestyle, dietary, or infectious trigger is implicated or expected.

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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. A **de novo *CCND2* variant** disrupts the C-terminal GSK-3β phosphodegron (Thr280) → **leads to** loss of the phosphorylation site that normally marks cyclin D2 for destruction.
2. Loss of Thr280 phosphorylation → **results in** failure of SCF/proteasome-mediated degradation → **cyclin D2 protein accumulates** (degradation-resistant). (Demonstrated in

vitro —

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P 24705253

3. Elevated cyclin D2 → **leads to** sustained assembly of **cyclin D2–CDK4/CDK6** complexes and prolonged kinase activity (overriding INK4/CIP-KIP inhibitors). (Biochemistry —

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P 16357141

4. Active CDK4/6 → **results in** hyperphosphorylation/inactivation of **retinoblastoma protein (pRb)** → **release of E2F** transcription factors → G1→S transition. (Inferred for neural progenitors from canonical D-cyclin biology — PMIDs 15684604, 16357141.)
5. In neural progenitors (radial glia), persistent G1/S drive → **leads to more cells remaining in cycle and fewer exiting to differentiate** (demonstrated by in utero electroporation of mutant CCND2 —

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P 24705253

6. Excess/prolonged progenitor proliferation → **results in brain overgrowth (megalencephaly)** and **disordered cortical architecture/lamination → polymicrogyria**; enlarged brain + altered CSF dynamics → **ventriculomegaly/hydrocephalus** (step partly inferred).
7. Convergent branch: **Upstream PI3K–AKT–mTOR activation** (PIK3CA/MCAP, PIK3R2/MPPH1, AKT3/MPPH2) → inhibits GSK-3β → **same cyclin D2 stabilization** → phenocopy.

> "cells from individuals with PIK3CA, PIK3R2 or AKT3 mutations showed similar CCND2 accumulation... cyclin D2 stabilization, caused by CCND2 mutation or PI3K-AKT activation, is a unifying mechanism in PI3K-AKT-related megalencephaly syndromes." ( )
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> P 24705253

8. Limb branch: PI3K–AKT signaling in limb buds (and, by inference, cyclin D2 dosage) contributes to **postaxial polydactyly** — "Our data highlight the central role of PI3K-AKT signaling in vascular, limb and brain development." ( )

P 22729224

Molecular pathways: PI3K–AKT–mTOR (KEGG hsa04151); Cell cycle (KEGG hsa04110); Rb–E2F/G1-S. **Cellular processes:** positive regulation of cell-cycle progression, reduced cell-cycle exit/differentiation of neural progenitors, altered neuronal migration (abnormal migration noted

clinically,

). P 38700464

Protein dysfunction: gain-of-function by stabilization/accumulation, not misfolding. **Metabolic link:** PI3K–AKT activation can cause **hypoketotic hypoinsulinaemic hypoglycaemia** in the broader pathway — relevant comorbidity to monitor (PMIDs 29883676, 28566443). **Immune involvement:** none. **Upstream vs downstream:** CCND2 is the **downstream effector/convergence node**; PIK3CA/PIK3R2/AKT3 are upstream.

Suggested ontology terms (CCND2 GO Biological Process annotations, verified via MyGene.info/GO): GO:0000082 (G1/S transition of mitotic cell cycle); GO:1900087 (positive regulation of G1/S transition of mitotic cell cycle); GO:2000045 (regulation of G1/S transition of mitotic cell cycle); GO:1901992 (positive regulation of mitotic cell cycle phase transition); GO:0008284 (positive regulation of cell population proliferation); **GO:0043066 (negative regulation of apoptotic process)** — relevant because reduced progenitor apoptosis compounds the overgrowth. Additional relevant: GO:0045737 (positive regulation of CDK activity); GO:0021846 (cell proliferation in forebrain). **Cell types (CL):** CL:0000031 neuroblast, CL:0000681 radial glial cell, CL:0000127 astrocyte, CL:0000120 cerebellar granule cell.

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7. Anatomical Structures Affected

Organ level (primary): Brain — cerebral cortex (perisylvian/opercular regions), white matter, ventricular system; cerebellum (granule/stellate neurons; tonsillar ectopia). **Body systems:** central nervous system (primary); musculoskeletal (limb — polydactyly); occasionally ocular (e.g., persistent hyperplastic primary vitreous reported in the spectrum,

). P 38008077

Secondary involvement: hydrocephalus-related raised intracranial pressure; epilepsy-related networks. **Tissue/cell level:** neuroepithelial/neural progenitor (radial glia) proliferative zones (ventricular/subventricular zones); cortical neurons; cerebellar granule and interneurons. **Subcellular (GO Cellular Component / UniProt P30279):** nucleus (GO:0005634, where cyclin D2–CDK acts on Rb/E2F); cytoplasm; nucleus membrane (UniProt-annotated localizations for G1/S-specific cyclin-D2); cyclin-dependent protein kinase holoenzyme complex (GO:0000307); cytoplasm also hosts the degradation machinery. **Localization / lateralization:** Brain involvement is **bilateral**; polymicrogyria is characteristically **bilateral perisylvian/**

symmetric. **UBERON:** UBERON:0000955 brain; UBERON:0000956 cerebral cortex; UBERON:0002037 cerebellum; UBERON:0002285 lateral ventricle; UBERON:0002544 digit (limb).

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8. Temporal Development

- **Onset:** Congenital/prenatal — megalencephaly, polymicrogyria, ventriculomegaly, and polydactyly are detectable in utero (prenatal ultrasound/MRI diagnoses reported, PMIDs 38008077, 39395260). **Pattern:** chronic/developmental (insidious structural, then infantile neurological onset).
- **Progression:** Megalencephaly is **progressive** ("progressive megalencephaly,"

  P 38700464

  ); ventriculomegaly may progress to hydrocephalus requiring intervention; cortical malformation is static once formed but its functional consequences (epilepsy, delay) evolve.
- **Course:** Chronic, lifelong; no spontaneous remission of structural disease. Seizures may be controllable (e.g., spasms resolving with topiramate in one case,

  P 39395260

) or drug-resistant.
- **Critical period:** Fetal corticogenesis (progenitor proliferation/neuronal migration) is the window in which the malformation is established; this is the mechanistic rationale/limitation for any disease-modifying therapy (post-natal therapy cannot reverse already-formed cortex).

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9. Inheritance and Population

- **Epidemiology:** Ultra-rare; MPPH (all genotypes) has **fewer than 100 reported cases** (

  P 39395260

  ); MPPH3 (*CCND2*) is a subset. No reliable prevalence/incidence figures exist (Orphanet "<1/1,000,000").

- **Inheritance: Autosomal dominant**, essentially always arising **de novo** (sporadic). Recurrence risk to siblings is low but non-zero due to possible **gonadal/germline mosaicism**.
- **Penetrance / expressivity:** Penetrance for the stabilizing variants is effectively **complete**; **expressivity is variable** (e.g., polydactyly present in some, absent in others; seizure severity varies).
- **Anticipation:** Not applicable (not a repeat-expansion disorder).
- **Founder effects / carrier frequency / consanguinity:** Not applicable — de novo dominant; no founder alleles, no carrier state, consanguinity not a factor.
- **Demographics:** No ethnic predilection reported; both sexes affected (no strong sex bias documented). Age distribution: pediatric (diagnosed prenatally to early childhood).

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10. Diagnostics

Imaging (central to diagnosis). Brain **MRI** is the key test: megalencephaly, **bilateral perisylvian polymicrogyria**, ventriculomegaly/hydrocephalus, ± cerebellar tonsillar ectopia, ± abnormal neuronal migration/white-matter changes. Prenatal **ultrasound/fetal MRI** can detect megalencephaly, abnormal gyration, and ventriculomegaly (PMIDs 38008077, 39395260). **EEG** for seizures/infantile spasms.

Genetic testing (confirmatory). - **Exome/genome sequencing (WES/WGS)** or **targeted NGS gene panels** covering the PI3K–AKT–mTOR megalencephaly genes (*CCND2*, *AKT3*, *PIK3R2*, *PIK3CA*) — the recommended first-line molecular approach. - **Single-gene / targeted *CCND2* sequencing** of the terminal exon/Thr280 region confirms MPPH3. - **Deep/mosaic-aware sequencing of affected tissue** may be needed when mosaicism is suspected (low variant allele fraction in blood) — "Sanger sequencing is ineffective for the detection of a causal variant because of the mosaic nature of these variants, whereas targeted next-generation sequencing technology offers a deeper sequencing coverage and allows the detection of low-level mosaicism."
(P 28502730)
- **Chromosomal microarray/karyotype:** typically normal; used to exclude CNVs (P 38700464).
- Mitochondrial/repeat-expansion testing: not indicated.

Biomarkers / labs: No specific circulating biomarker. **Monitor blood glucose** given the pathway's association with hypoinsulinaemic, hypoketotic **hypoglycaemia** (PMIDs 29883676, 28566443). Metabolic screening (tandem MS) is typically normal (P 38700464).

Clinical criteria / differential diagnosis. Diagnosis = megalencephaly + cortical malformation (polymicrogyria) ± polydactyly ± hydrocephalus, confirmed by a pathogenic PI3K–AKT–pathway variant. **Differentials:** MCAP (has capillary malformation/somatic overgrowth, usually *PIK3CA* mosaic); MPPH1 (*PIK3R2*), MPPH2 (*AKT3*); other megalencephaly/overgrowth syndromes (e.g., PTEN, Sotos, hemimegalencephaly); isolated polymicrogyria. Distinguishing feature of MPPH vs MCAP: **absence of capillary/cutaneous vascular malformation and segmental somatic overgrowth** in MPPH.

Screening: No population newborn screening. Cascade testing of parents is generally uninformative (de novo) but relevant for recurrence counseling (mosaicism).

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11. Outcome / Prognosis

- **Survival/mortality:** No formal survival statistics; severe cases with marked hydrocephalus, drug-resistant epilepsy, or complications carry risk, but many survive into childhood/adulthood with disability. A rare report documents **medulloblastoma** arising in an MPPH patient (P 33634562), consistent with cyclin D2/PI3K-pathway oncogenic potential — a possible tumor-surveillance consideration.
- **Morbidity/function:** Dominated by **moderate-to-severe/profound intellectual disability and motor impairment**, epilepsy, and hydrocephalus-related morbidity; many patients are nonverbal with hypotonia (P 39395260).
- **Disease course:** Chronic, lifelong; complications include progressive hydrocephalus (may need shunt/ETV), refractory seizures, and feeding/airway issues in severe cases.

- **Prognostic factors:** Severity of megalencephaly/cortical malformation, hydrocephalus requiring diversion, epilepsy control, and presence of complications (e.g., IVH). No validated molecular prognostic biomarker.

12. Treatment

No disease-specific curative therapy exists; management is **multidisciplinary and largely supportive**.

- **Neurosurgical:** CSF diversion (**ventriculoperitoneal shunt** or **endoscopic third ventriculostomy**) for hydrocephalus; epilepsy surgery in selected drug-resistant cases.
- **Antiseizure medication:** Standard AEDs; infantile spasms treated per protocol. In one *CCND2* case, high-dose prednisone failed but **topiramate** achieved spasm control (P 39395260). (NCIT examples: C66883 Topiramate; C716 Prednisone.)
- **Rehabilitation/supportive:** Physical, occupational, and speech therapy; developmental/educational support; orthopedic correction of polydactyly.
- **Metabolic monitoring:** Surveillance for **hypoglycaemia** (pathway-associated); treat as indicated (PMIDs 29883676, 28566443).
- **Targeted/experimental (pathway-directed, mostly investigational for brain disease):** Because MPPH converges on PI3K–AKT–mTOR/cyclin D2, pathway inhibitors are mechanistically rational:
- **mTOR inhibitors (sirolimus/rapamycin)** — partial efficacy in overgrowth phenotypes but durable responses uncommon and recurrence after discontinuation observed (P 42747160). NCIT C1212 Sirolimus.
- **PI3Kα inhibitors (alpelisib; next-gen mutant-selective zovegalisib/RLY-2608)** — established in PIK3CA-related overgrowth spectrum (PMIDs 42747160, 42875408); not established for fixed CNS malformation. NCIT C123846 Alpelisib.
- **CDK4/6 inhibitors** — a conceptually attractive node directly downstream of cyclin D2, but not clinically tested in MPPH.

- **Key limitation:** the cortical malformation is established prenatally, so postnatal pathway inhibition is unlikely to reverse structural disease; benefit (if any) would target overgrowth/progression, epilepsy, or tumor risk. Gene/RNA therapies: none.

13. Prevention

- **Primary prevention:** Not possible (de novo mutation). **Genetic counseling** emphasizes low but non-zero recurrence risk (germline mosaicism).
- **Secondary prevention / prenatal: Prenatal ultrasound/fetal MRI** can detect megalencephaly, abnormal gyration, ventriculomegaly, enabling early diagnosis and counseling (PMIDs 38008077, 39395260). **Prenatal/preimplantation genetic testing** is feasible if a familial variant is known (e.g., proven parental mosaicism).
- **Tertiary prevention:** Prevent complications — hydrocephalus management, seizure control, glucose monitoring, developmental intervention.
- **Public health / immunization / behavioral:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs (HomoloGene, verified):** *CCND2* is broadly conserved. Ortholog NCBI Gene IDs — human 894 (taxon 9606); mouse *Ccnd2* **12444** (10090); rat **64033** (10116); zebrafish **799608** (*Danio rerio*, 7955); chicken 374047 (9031); dog 611782 (9615); cow 615414 (9913); chimpanzee 747123 (9598); *Xenopus tropicalis* 100038071 (8364); *Drosophila* ortholog (CycD) 32551 (7227). Conservation from invertebrates to mammals underscores the ancient role of D-cyclins in cell-cycle control.
- **Natural disease in animals:** No recognized naturally occurring *CCND2* megalencephaly syndrome in companion animals/wildlife (not in OMIA as a spontaneous disease). MPPH3 is studied via engineered models, not natural animal disease.
- **Comparative biology:** Cyclin D2's role in neural (especially cerebellar granule-cell) proliferation is evolutionarily conserved, supporting cross-species mechanistic relevance.
- **Zoonotic potential:** None (non-infectious genetic disorder).

15. Model Organisms

- **Mouse — *Ccnd2* knockout (loss of function):** "The loss of this cyclin primarily affected two neuronal populations: granule cell number was reduced and stellate interneurons were nearly absent" with reduced cerebellar neurogenesis and increased apoptosis (Huard et al. 1999, P 10101126).
  This models the **inverse (hypoplasia/microcephaly)** pole, validating cyclin D2 as a dose-dependent brain-growth regulator. Thyroid-hormone studies confirm cyclin D2 drives cerebellar EGL neuroblast proliferation/anti-apoptosis (P 12591240).
- **In utero electroporation of mutant (stabilized) CCND2 in embryonic mouse cortex:** recapitulates the **gain-of-function** mechanism — "produced more proliferating transfected progenitors and a smaller fraction of progenitors exiting the cell cycle compared to cells electroporated with wild-type CCND2" (P 24705253).
  This is the most direct in vivo model of the MPPH3 mechanism.
- **Activated-AKT3 mouse:** shows elevated brain cyclin D2, linking upstream pathway activation to the effector (P 24705253).
- **Model types available:** knockout (loss-of-function), in utero electroporation/ overexpression (gain-of-function), and transgenic pathway-activation models. **Recapitulation:** overexpression/electroporation models reproduce excess progenitor proliferation; full syndromic replication (polydactyly, hydrocephalus) is incomplete. **Limitations:** no single model reproduces the entire human tetrad; timing/dosage-sensitive.
- **In vitro:** patient-derived cells show cyclin D2 accumulation; biochemical assays confirm degradation resistance (P 24705253).
- **Resources:** MGI (mouse *Ccnd2*), IMPC/KOMP for knockout lines.

Supported vs Refuted Hypotheses

Supported: 1. MPPH3 is caused by de novo *CCND2* gain-of-function variants that stabilize cyclin D2 against proteasomal degradation (P 24705253). 2. The mechanism is excess/prolonged neural-progenitor proliferation via cyclin D2–CDK4/6–Rb–E2F, converging with upstream PI3K–AKT–mTOR activation (PMIDs 24705253, 16357141, 22729224). 3. Cyclin D2 dosage controls brain size bidirectionally — loss → microcephaly, gain → megalencephaly (PMIDs 10101126, 34087052).

Refuted / ruled out: - Environmental, infectious, or metabolic primary causation (no evidence; disease is monogenic de novo). - An intrinsic enzymatic gain (the defect is protein **abundance/stability**, not altered catalytic activity).

Limitations

- Evidence rests on rare case reports and a small number of functional studies; no large cohorts, registries, or omics datasets specific to MPPH3.
- Genotype–phenotype correlations (e.g., variable polydactyly) remain unexplained.
- No clinical trial data for targeted therapy in MPPH3 specifically; pathway-inhibitor evidence is extrapolated from PIK3CA-related overgrowth.

Future Directions

- Prospective natural-history/registry studies of *CCND2*-specific MPPH3.
- Trials of CDK4/6 or PI3K–AKT–mTOR inhibitors for progressive/overgrowth or epilepsy endpoints, defining therapeutic windows.
- Tumor-risk surveillance guidance given isolated medulloblastoma report (P 33634562).

Key references (PMID)

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